MD SANZID BIN HOSSAIN

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RESEARCH SUMMARY

My research primarily focuses on advanced AI-based approaches, utilizing wearables for more practical, cost-effective, and accurate human gait analysis. I have published several high-impact journals and presented at major conferences. My future goal is to apply my AI expertise to in-depth biomechanical analysis, assistive device control and evaluation through novel methods, and ubiquitous clinical assessment.

RESEARCH EXPERIENCE

- **Postdoctoral Scholar**, University of Central Florida, Orlando, FL — *August 2024- Current*
  Department of Clinical Science, College of Medicine
  Advisor: Dr. Dexter Hadley
- **Graduate Research Assistant**, University of Central Florida, Orlando, FL — *August 2020- July 2024*
  Department of Electrical and Computer Engineering
  Advisor: Dr. Hwan Choi and Dr. Zhishan Guo

EDUCATION

University of Central Florida (UCF), Orlando, FL — *August 2024*
PhD in Computer Engineering, Department of ECE

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh — *September 2017*
Bachelor of Science, Department of Electrical and Electronics Engineering

SELECTED AWARDS AND HONORS

- NSF Student Travel Award, IEEE/ACM CHASE — 2022
- ORC Doctoral Fellow, University of Central Florida — 2019
- UCF Presentation Fellowship — 2022, 2024

RESEARCH PAPER PUBLICATIONS

Major Research Topics

- Wearables
- Artificial Intelligence (AI) and Deep Learning
- Human Gait Monitoring
- Biomechanics

PhD Dissertation

"Towards Sparse IMU Sensor-Based Estimation of Walking Kinematics, Joint Moments, and Ground Reaction Forces in Multiple Locomotion Modes via Deep Learning."

Journal Publications

[J8] Md Moniruzzaman, Zhaozheng Yin, **Md Sanzid Bin Hossain**, Hwan Choi, and Zhishan Guo, "Wearable Motion Capture: Reconstructing and Predicting 3D Human Poses From Wearable Sensors," in *IEEE Journal of Biomedical and Health Informatics*, vol. 27, no. 11, pp. 5345–5356, 2023.

[J7] **Md Sanzid Bin Hossain**, Zhishan Guo, and Hwan Choi, “Estimation of Lower Extremity Joint Moments and 3D Ground Reaction Forces Using IMU Sensors in Multiple Walking Conditions: A Deep Learning Approach,” in *IEEE Journal of Biomedical and Health Informatics*, 2023.

[J6] **Md Sanzid Bin Hossain**, Joseph Dranetz, Hwan Choi, and Zhishan Guo, “DeepBBWAENet: A CNN-RNN Based Deep Superlearner for Estimating Lower Extremity Sagittal Plane Joint Kinematics Using Shoe-Mounted IMU Sensors in Daily Living,” in *IEEE Journal of Biomedical and Health Informatics*, vol. 26, no. 8, pp. 3906–3917, 2022.

[J5] **Md Sanzid Bin Hossain**, Yelena Piazza, Jacob Braun, Anthony Bilic, Michael Hsieh, Samir Fouissi, Alexander Borowsky, Hatem Kaseb, Chaithanya Renduchintala, Amoy Fraser, Britney-Ann Wray, Chen Chen, Liqiang Wang, Mujtaba Husain, and Dexter Hadley, “UCF-MultiOrgan-Path: A Benchmark Dataset of Histopathologic Images for Deep Learning-Based Organ Classification”, **Under submission**

[J4] **Md Sanzid Bin Hossain**, Dexter Hadley, Zhishan Guo, and Hwan Choi, “Lightweight and Efficient Kinetics Estimation with Sparse IMUs via Multi-Modal Sensor Distillation”, **Under submission**

[J3] **Md Sanzid Bin Hossain**, Hwan Choi, Zhishan Guo, Hyunjun Shin, and Dexter Hadley, “Smartphone-Video-Based Estimation of Knee Joint Kinetics and Ground Reaction Forces via Multi-modal Fusion and Knowledge Distillation”, **Under submission**

[J2] Oliver Fritsche, Steven Camacho, **Md Sanzid Bin Hossain**, Tyler Halpenny, Carlos Archniegas, Joseph Dranetz, Dexter Hadley, Zhishan Guo, and Hwan Choi, ”ULTRA-MoCap: A Multimodal IMU and sEMG Dataset for Deep Learning-Based Upper Body Joint Kinematic Analysis”, **Under submission** 2025.

[J1] **Md Sanzid Bin Hossain**, Dexter Hadley, Zhishan Guo, and Hwan Choi, “Bridging the Data Gap: Cross-Dataset Knowledge Transfer Across Heterogeneous Sensors for Enhanced Kinetics Estimation”, **Under preparation** 2025.

Conference Proceedings

[C7] **Md Sanzid Bin Hossain**, Zhishan Guo, Ning Sui, and Hwan Choi, “Predicting Lower Extremity Joint Kinematics Using Multi-Modal Data in the Lab and Outdoor Environment,” in *57th Hawaii International Conference on System Sciences*, 2024.

[C6] **Md Sanzid Bin Hossain**, Zhishan Guo, and Hwan Choi, “Estimation of Hip, Knee, and Ankle Joint Moment Using a Single IMU Sensor on Foot Via Deep Learning,” in *2022 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE)*, pp. 25–33, 2022.

[C5] **Md Sanzid Bin Hossain**, Hwan Choi, and Zhishan Guo, “Estimating Lower Extremity Joint Angles During Gait Using Reduced Number of Sensors Count Via Deep Learning,” in *Fourteenth International Conference on Digital Image Processing (ICDIP 2022)*, vol. 12342, pp. 1116–1123, 2022.

[C4] **Md Sanzid Bin Hossain**, Md Shazid Islam, Md Saad Ul Haque, and Md Saydur Rahman, “Gait Phase Classification from sEMG in Multiple Locomotion Mode Using Deep Learning,” in *9th International Congress on Information and Communication Technology*, 2024.

[C3] Jayanta Dey, **Md Sanzid Bin Hossain**, and Mohammad Ariful Haque, “An Ensemble SVM-Based Approach for Voice Activity Detection,” in *2018 10th International Conference on Electrical and Computer Engineering (ICECE)*, pp. 297–300, 2018.

[C2] Md Shazid Islam, Md Saydur Rahman, Md Saad Ul Haque, Farhana Akter Tumpa, **Md Sanzid Bin Hossain**, and Abul Al Arabi, “Location Agnostic Adaptive Rain Precipitation Prediction Using Deep Learning,” in *2023 IEEE 9th International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE)*, pp. 148–153, 2023.

[C1] Md Saydur Rahman, Farhana Akter Tumpa, Md Shazid Islam, Abul Al Arabi, **Md Sanzid Bin Hossain**, and Md Saad Ul Haque, “Comparative Evaluation of Weather Forecasting Using Machine Learning Models,” in *2023 26th International Conference on Computer and Information Technology (ICCIT)*, pp. 1–6, 2023.

Conference Abstract

[CA5] **Md Sanzid Bin Hossain**, Zhishan Guo, and Hwan Choi, "Domain Adaptation Technique to Transfer Knowledge Among Different Datasets for Improved Kinetics Estimation," in *Proceedings of the Gait and Clinical Movement Analysis Society Annual Meeting (GCMAS)*, 2024.

[CA4] **Md Sanzid Bin Hossain**, Zhishan Guo, and Hwan Choi, "Sensor and Cross-Modal Knowledge Distillation for Kinetics Estimation," in *Proceedings of the 46th Meetings of the American Society of Biomechanics (ASB)*, 2023.

[CA3] **Md Sanzid Bin Hossain**, Zhishan Guo, and Hwan Choi, "Estimating Ground Reaction Forces (GRF) and Lower Extremity Joint Moment in Multiple Walking Environments Using Deep Learning," in *Proceedings of the Gait and Clinical Movement Analysis Society Annual Meeting (GCMAS)*, 2022.

[CA2] **Md Sanzid Bin Hossain**, Joseph Dranetz, Hwan Choi, and Zhishan Guo, "An Ensemble Machine Learning Approach for the Estimation of Lower Extremity Kinematics Using Shoe-Mounted IMU Sensors," in *Proceedings of the Gait and Clinical Movement Analysis Society Annual Meeting (GCMAS)*, 2021.

[CA1] **Md Sanzid Bin Hossain**, Youngho Lee, Junghwa Hong, Hwan Choi, and Zhishan Guo, "Predicting Lower Limb 3D Kinematics During Gait Using Reduced Number of Wearable Sensors Via Deep Learning," in *Proceedings of the 44th Meetings of the American Society of Biomechanics (ASB)*, 2020.

GRANT PREPARATION

[G1] [NIH R01], "Wearable and Ubiquitous Motion Sensing and Analysis with Physics-Informed Machine Learning for Movement-Disorder Patients" Role: Other significant consultant (conceptualized and wrote the physics-informed deep learning algorithm portion). Funding amount: $2,136,320, Pending. 2025

TALKS/POSTER PRESENTATIONS

[T8] **[Podium Presentation]** 9th International Congress on Information and Communication Technology 2024
[T7] **[Podium Presentation]** Gait and Clinical Movement Analysis Society Annual Meeting (GCMAS) 2024
[T6] **[Poster Presentation]** 46th Meetings of the American Society of Biomechanics (ASB) 2023
[T5] **[Podium Presentation]** Gait and Clinical Movement Analysis Society Annual Meeting (GCMAS) 2022
[T4] **[Podium Presentation]** IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE) 2022
[T3] **[Podium Presentation]** Fourteenth International Conference on Digital Image Processing (ICDIP) 2022
[T2] **[Poster Presentation]** Gait and Clinical Movement Analysis Society Annual Meeting (GCMAS) 2021
[T1] **[Poster Presentation]** 46th Meetings of the American Society of Biomechanics (ASB) 2020

RESEARCH PROJECTS

[P4] Whole Slide Image (WSI) Cadaver Tissue Patch Preparation and Classification
- Curated and prepared 1,700 WSIs from various organs for further analysis.
- Created patches from these WSIs to establish a benchmark dataset for classifying 15 different organs.

[P3] Smartphone-Video Based Human Kinetics Estimation
- Developed an innovative method for estimating knee adduction moment (KAM), knee flexion moment (KFM), and 3D ground reaction forces (GRFs) using only smartphone video data.
- The method employs a two-step knowledge distillation and multi-modal fusion technique to achieve high accuracy in kinetics estimation.

[P2] Sparse IMU Sensor-Based Joint Angles, Joint Moments, and GRFs Estimation
- Proposed novel methods for estimating joint angles, joint moments, and ground reaction forces (GRFs) using a sparse configuration of inertial measurement unit (IMU) sensors.

- Leveraged a new technique called sensor distillation and multi-modal fusion to enhance the accuracy and reliability of the estimations, significantly outperforming state-of-the-art deep learning models while remaining computationally lightweight.

[P1] Prediction and Reconstruction of 3D Human Pose Using IMUs and Wearable Cameras
- Collected data from 20 subjects using Vicon Motion Capture, Delsys IMUs, and two shank-mounted egocentric GoPro cameras, with subsequent processing in OpenSim.
- Introduced an innovative approach that integrates whole-body IMUs with wearable camera features to accurately reconstruct current 3D poses and predict future movements.

TEACHING EXPERIENCE (TA)

- EEL 4768: Computer Architecture — Summer'22
- EEL 4742C: Embedded Systems Lab — Fall'22
- EEL3801C: Computer Organization — Summer'22
- EEL 3021: Introduction to Applied Randomness for Engineers — Fall'22
- EEE 4775/EEL 5862: Real-Time-Systems — Fall'22, Spring'23
- EEL 3552C: Signal Analysis and Analog Communication — Spring'23
- EEE 3307C: Electronics I — Fall'22, Spring'23

PROFESSIONAL SERVICE

- Reviewer, AAAI Conference on Artificial Intelligence (AAAI) — 2023
- Reviewer, Real-Time and Network Systems (RTNS) — 2022
- Reviewer, IEEE Real-Time and Embedded Technology and Applications Symposium(RTAS) — 2021
- Reviewer, IEEE Computer Society Signature Conference on Computers, Software, and Applications (COMPSAC) — 2023
- Reviewer, International Conference on Neural Information Processing (ICONIP) — 2021
- Reviewer, International Conference on High Performance Big Data and Intelligent Systems (HDIS) — 2022
- Reviewer, ACM International Conference on Web Search and Data Mining (WSDM) — 2020
- Reviewer, International Conference on Information Society and Technology (ICIST) — 2020
- Reviewer, IEEE International Conference on Embedded and Real-Time Computing Systems and Application (RTCSA) — 2022
- Reviewer, Computer Methods in Biomechanics and Biomedical Engineering — 2023
- Secondary Reviewer, IEEE Transactions on Neural Systems and Rehabilitation Engineering — 2024

SKILLS

- *Programming Languages*: C, Python, MATLAB.
- *Hardware/Instrumentation*:
  - Expertise in human motion analysis using **Vicon motion capture system** for biomechanical and clinical research.
  - Proficient in integrating **Delsys EMG and IMU systems**, **AMTI force plates**, and **instrumented treadmills** for data collection and experimental studies.

- Extensive experience in using **OpenSim** for musculoskeletal modeling and gait analysis.

- *Operating Systems and Software*: Linux, Windows, Office Software, LaTeX, Google Cloud Platforms (GCP).

MENTORING EXPERIENCE

- **Anirudh Kaluri**, Research Assistant, MS student, North Carolina State University
- **Carlos Arciniegas, Oliver Fritsche, Steven Camacho, Tyler Halfpenny**, Undergraduate Senior Design Project, University of Central Florida
- **Yi Hong Jong, Pranav Sattiraju, Samir Fouissi**, Undergraduate Research Assistant, University of Central Florida

TRAINING

- Information Security Awareness Training (UCF) 2024
- Fraud Awareness Training (UCF) 2024
- Biomedical Responsible Conduct of Research (UCF) 2023
- Laboratory Hazardous Waste Handling and Processing (UCF) 2023
- Graduate Teaching Assistant Training (UCF) 2021

REFERENCES

- **Dr. Hwan Choi**
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  Assistant Professor, Mechanical and Aerospace Engineering (MAE)
  University of Central Florida
  Phone: 407-266-7130
  Email: hwan.choi@ucf.edu
- **Dr. Zhishan Guo**
  PhD Advisor
  Associate Professor, Department of Computer Science
  North Carolina State University
  Phone: 919-515-3962
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- **Dr. Dexter Hadley**
  Postdoctoral Advisor
  Assistant Professor, Department of Clinical Science, College of Medicine
  University of Central Florida
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